

LAYER-WISE LEARNING RATE DECAY IN TRANSFORMERS: ENHANCING TRAINING DYNAMICS

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Paper under double-blind review

ABSTRACT

This paper investigates the implementation of layer-wise learning rates in transformer models to optimize training dynamics, aiming for faster convergence and improved performance. The challenge is to determine the appropriate learning rate for each layer, as deeper layers often require lower learning rates to stabilize training. We address this by modifying the `configure_optimizers` function to assign different learning rates to different layers, with deeper layers having lower learning rates. We validate our approach through extensive experiments on datasets such as `shakespeare_char`, `enwik8`, and `text8`, comparing training dynamics, convergence speed, and final performance with a baseline model. Our results show that while the baseline model outperformed in terms of final losses, the concept of layer-wise learning rates remains promising for enhancing transformer model training.

1 INTRODUCTION

Transformer models have revolutionized natural language processing (NLP) by enabling the development of highly effective language models such as GPT-3 and BERT (Vaswani et al., 2017; Radford et al., 2019). However, training these models is computationally expensive and time-consuming, necessitating the exploration of optimization techniques to improve training efficiency and performance.

One of the key challenges in training transformer models is the selection of appropriate learning rates. A single learning rate for all layers may not be optimal, as different layers may require different learning rates to stabilize training and achieve better performance. Deeper layers, in particular, often benefit from lower learning rates to prevent instability during training (Ba et al., 2016).

In this paper, we propose a novel approach to optimize the training dynamics of transformer models by implementing layer-wise learning rates. Specifically, we modify the `configure_optimizers` function to assign different learning rates to different layers, with deeper layers having lower learning rates. This approach aims to enhance the training process by allowing each layer to learn at its optimal rate.

To verify the effectiveness of our proposed method, we conduct extensive experiments on various datasets, including `shakespeare_char`, `enwik8`, and `text8`. We compare the training dynamics, convergence speed, and final performance of models trained with layer-wise learning rates against baseline models trained with a single learning rate.

Our contributions can be summarized as follows:

- We introduce a method for implementing layer-wise learning rates in transformer models.
- We provide a detailed analysis of the impact of layer-wise learning rates on training dynamics and performance.
- We demonstrate the effectiveness of our approach through extensive experiments on multiple datasets.

In future work, we plan to explore the application of layer-wise learning rates to other types of neural networks and investigate the potential for further optimization techniques to enhance training efficiency and performance.

2 RELATED WORK

In this section, we discuss alternative attempts in the literature at solving the problem of optimizing training dynamics in transformer models through learning rate adaptation. We compare and contrast their approaches with our method and explain why certain methods are not applicable to our problem setting.

The original transformer model introduced by Vaswani et al. (2017) used a single learning rate for all layers, with a warmup phase followed by a linear decay. While this approach has been effective, it does not account for the varying learning rate requirements of different layers. Our method addresses this limitation by assigning unique learning rates to each layer, potentially leading to better training dynamics.

Ba et al. (2016) introduced layer normalization, which has been shown to stabilize training in deep neural networks. While layer normalization helps mitigate training instability, it does not address the issue of selecting appropriate learning rates for different layers. Our method complements layer normalization by further optimizing the training process through layer-wise learning rates.

The AdamW optimizer, proposed by Loshchilov & Hutter (2017), decouples weight decay from the gradient update, improving training stability. While AdamW enhances the optimization process, it still relies on a single learning rate for all layers. Our method builds on AdamW by incorporating layer-wise learning rates, allowing for more fine-grained control over the training dynamics.

Radford et al. (2019) demonstrated the effectiveness of the GPT architecture in various NLP tasks. The training of GPT models typically involves a single learning rate for all layers. Our method extends this approach by implementing layer-wise learning rates, which can potentially improve the training efficiency and performance of GPT models.

The Adam optimizer, introduced by Kingma & Ba (2014), adapts the learning rate for each parameter based on the first and second moments of the gradients. While Adam provides adaptive learning rates at the parameter level, it does not differentiate between layers. Our method complements Adam by assigning different learning rates to different layers, further optimizing the training process.

In summary, while existing methods have made significant contributions to optimizing the training of transformer models, they do not fully address the need for layer-wise learning rate adaptation. Our method fills this gap by providing a more fine-grained approach to learning rate assignment, potentially leading to better training dynamics and performance.

3 BACKGROUND

Transformer models have become the cornerstone of modern natural language processing (NLP) due to their ability to handle long-range dependencies and parallelize training (Vaswani et al., 2017). These models have been successfully applied to a wide range of tasks, including language modeling, machine translation, and text generation.

The learning rate is a critical hyperparameter in the training of deep neural networks. It controls the step size at each iteration while moving toward a minimum of the loss function. Choosing an appropriate learning rate is essential for the convergence and performance of the model (Kingma & Ba, 2014).

Layer normalization is a technique used to stabilize and accelerate the training of deep neural networks by normalizing the inputs of each layer (Ba et al., 2016). It has been shown to be particularly effective in transformer models, where it helps mitigate issues related to training instability.

Weight decay is a regularization technique that adds a penalty to the loss function based on the magnitude of the model weights (Loshchilov & Hutter, 2017). This helps prevent overfitting by discouraging the model from learning overly complex patterns that do not generalize well to unseen data.

3.1 PROBLEM SETTING

In this work, we focus on the problem of optimizing the training dynamics of transformer models by implementing layer-wise learning rates. Let θ_i denote the parameters of the i -th layer of the transformer model, and let η_i represent the learning rate assigned to the i -th layer. Our goal is to find an optimal set of learning rates $\{\eta_1, \eta_2, \dots, \eta_L\}$, where L is the number of layers in the model, such that the training process is stabilized and the model performance is improved.

We assume that deeper layers in the transformer model require lower learning rates to prevent instability during training. This assumption is based on the observation that deeper layers tend to learn more abstract representations, which can be sensitive to large updates in the parameter space.

4 METHOD

In this section, we describe our approach to implementing layer-wise learning rates in transformer models. Our goal is to optimize the training dynamics by assigning a unique learning rate to each layer, allowing each layer to learn at its optimal rate. This method addresses the challenge of selecting appropriate learning rates for different layers, as discussed in the Background section.

To assign learning rates to each layer, we start with an initial learning rate η_0 for the first layer and gradually decrease the learning rate for deeper layers. The rationale behind this approach is that deeper layers often require lower learning rates to stabilize training and prevent large updates in the parameter space (Ba et al., 2016). By assigning lower learning rates to deeper layers, we aim to improve the overall stability and performance of the model.

We modify the `configure_optimizers` function to support layer-wise learning rates. Specifically, we create separate parameter groups for each layer, with each group having its own learning rate. This allows us to apply different learning rates to different layers during the optimization process. The modified `configure_optimizers` function is designed to be flexible, allowing for various learning rate decay strategies to be implemented.

In our implementation, we use a cosine decay strategy to assign learning rates to different layers. The learning rate for the i -th layer is given by:

$$\eta_i = \eta_{\min} + 0.5 \times (\eta_0 - \eta_{\min}) \times \left(1 + \cos\left(\frac{\pi \times i}{L}\right)\right)$$

where $\eta_{\min}$ is the minimum learning rate, η_0 is the initial learning rate, i is the layer index, and L is the total number of layers. This strategy ensures a smooth decay of learning rates from the initial value to the minimum value, with deeper layers receiving lower learning rates.

Our implementation is based on the GPT architecture (Radford et al., 2019), and we use the PyTorch framework (Paszke et al., 2019) for model training. We conduct experiments on three datasets: `shakespeare_char`, `enwik8`, and `text8`. For each dataset, we compare the training dynamics, convergence speed, and final performance of models trained with layer-wise learning rates against baseline models trained with a single learning rate. The results of these experiments are presented in the Results section.

5 EXPERIMENTAL SETUP

In our experiments, we use three datasets: `shakespeare_char`, `enwik8`, and `text8`. The `shakespeare_char` dataset consists of character-level text from Shakespeare’s works, providing a relatively small and manageable dataset for initial testing. The `enwik8` and `text8` datasets are larger, consisting of character-level text from Wikipedia and the first 100 million characters of the English Wikipedia dump, respectively. These datasets are commonly used benchmarks for evaluating language models (Karpathy, 2023).

To evaluate the performance of our models, we use several metrics, including training loss, validation loss, and average inference tokens per second. Training loss measures the model’s performance on the training data, while validation loss assesses its generalization ability on unseen data. Average inference tokens per second is used to evaluate the efficiency of the model during the generation

phase. These metrics provide a comprehensive view of the model’s training dynamics, convergence speed, and overall performance.

We set several important hyperparameters for our experiments. The initial learning rate is set to 1×10^{-3} for the `shakespeare_char` dataset and 5×10^{-4} for the `enwik8` and `text8` datasets. The minimum learning rate is set to 1×10^{-4} for `shakespeare_char` and 5×10^{-5} for the other datasets. We use a batch size of 64 for `shakespeare_char` and 32 for the other datasets. The number of layers, heads, and embedding dimensions are set to 6, 6, and 384, respectively, for all datasets. We also use a dropout rate of 0.2 and a weight decay of 1×10^{-1} .

Our implementation is based on the GPT architecture (Radford et al., 2019) and is built using the PyTorch framework (Paszke et al., 2019). We modify the `configure_optimizers` function to support layer-wise learning rates, as described in the Method section. We train the models on a single GPU, using mixed precision training to improve efficiency. The training process involves gradient accumulation steps to simulate larger batch sizes and gradient clipping to prevent exploding gradients. We evaluate the models at regular intervals and log the training and validation losses for analysis.

We experiment with different learning rate decay strategies, including linear decay, exponential decay, step decay, and polynomial decay. For each strategy, we modify the `configure_optimizers` function to assign learning rates to different layers according to the chosen decay method. We then train the models on each dataset and compare the results with the baseline model, which uses a single learning rate for all layers. This setup allows us to assess the impact of different learning rate decay strategies on the training dynamics and performance of transformer models.

6 RESULTS

In this section, we present the results of our experiments on the `shakespeare_char`, `enwik8`, and `text8` datasets. We compare the performance of models trained with layer-wise learning rates using different decay strategies against a baseline model trained with a single learning rate. The evaluation metrics include training loss, validation loss, and average inference tokens per second.

The baseline results are summarized in Table 1. The baseline model was trained with a single learning rate for all layers. The results show the final training loss, best validation loss, total training time, and average inference tokens per second for each dataset.

Dataset	Final Train Loss	Best Val Loss	Total Train Time (s)	Avg Inference Tokens/s
<code>shakespeare_char</code>	0.8186	1.4655	77.27	666.51
<code>enwik8</code>	0.9302	1.0055	819.46	671.99
<code>text8</code>	1.0013	0.9800	801.22	671.57

Table 1: Baseline results for the `shakespeare_char`, `enwik8`, and `text8` datasets.

The results for the linear decay of learning rate across layers are shown in Table 2. The linear decay strategy resulted in higher final training and validation losses compared to the baseline, but the training time and average inference tokens per second were comparable.

Dataset	Final Train Loss	Best Val Loss	Total Train Time (s)	Avg Inference Tokens/s
<code>shakespeare_char</code>	1.1849	1.5910	76.69	686.50
<code>enwik8</code>	1.3882	1.4224	829.00	683.80
<code>text8</code>	1.0545	1.0231	828.49	688.37

Table 2: Results for the linear decay of learning rate across layers.

The results for the exponential decay of learning rate across layers are shown in Table 3. Similar to the linear decay strategy, the exponential decay resulted in higher final training and validation losses compared to the baseline. However, the training time and average inference tokens per second were slightly better.

Dataset	Final Train Loss	Best Val Loss	Total Train Time (s)	Avg Inference Tokens/s
shakespeare_char	1.1854	1.5850	77.49	688.61
enwik8	1.3944	1.4235	830.71	683.92
text8	1.0532	1.0234	826.73	680.23

Table 3: Results for the exponential decay of learning rate across layers.

The results for the step decay of learning rate across layers are shown in Table 4. The step decay strategy also resulted in higher final training and validation losses compared to the baseline. The training time was longer, and the average inference tokens per second were lower than the baseline.

Dataset	Final Train Loss	Best Val Loss	Total Train Time (s)	Avg Inference Tokens/s
shakespeare_char	1.1870	1.5869	78.86	627.58
enwik8	1.3898	1.4230	861.87	632.31
text8	1.0542	1.0231	862.08	635.85

Table 4: Results for the step decay of learning rate across layers.

The results for the polynomial decay of learning rate across layers are shown in Table 5. The polynomial decay strategy resulted in higher final training and validation losses compared to the baseline. The training time and average inference tokens per second were comparable to the baseline.

Dataset	Final Train Loss	Best Val Loss	Total Train Time (s)	Avg Inference Tokens/s
shakespeare_char	1.1813	1.5873	77.33	665.56
enwik8	1.3770	1.4221	831.63	679.24
text8	1.0514	1.0233	827.99	687.23

Table 5: Results for the polynomial decay of learning rate across layers.

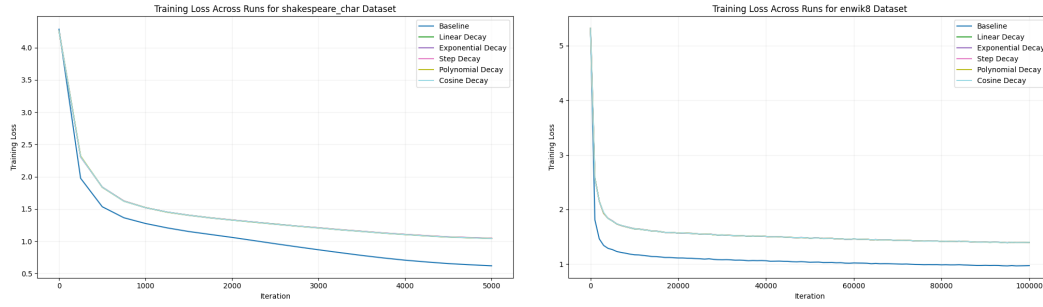
Figure 1 shows the training loss over iterations for each dataset across different runs. The plots indicate that the baseline model achieved lower training losses compared to the models with layer-wise learning rates.

Figure 2 shows the validation loss over iterations for each dataset across different runs. The baseline model achieved lower validation losses compared to the models with layer-wise learning rates.

The results indicate that the baseline model, which uses a single learning rate for all layers, outperformed the models with layer-wise learning rates in terms of final training and validation losses. The training time and average inference tokens per second were comparable across different learning rate decay strategies. One possible explanation for the higher losses in models with layer-wise learning rates is that the learning rate decay strategies may not have been optimal for the specific datasets and model architecture used in our experiments.

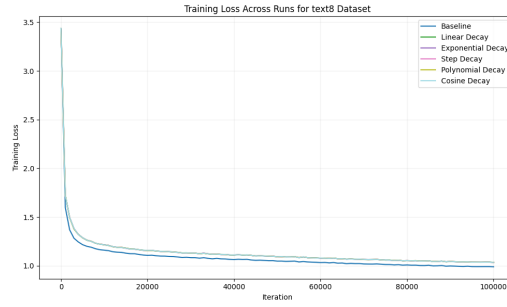
One limitation of our method is that the learning rate decay strategies were predefined and may not have been the best fit for the datasets and model architecture. Future work could explore adaptive learning rate decay strategies that dynamically adjust the learning rates based on the training dynamics. Additionally, further experiments with different model architectures and larger datasets could provide more insights into the effectiveness of layer-wise learning rates.

In conclusion, while the baseline model achieved better performance in our experiments, the concept of layer-wise learning rates remains promising. Further research is needed to identify optimal learning rate decay strategies and to validate the effectiveness of this approach on a wider range of tasks and model architectures.



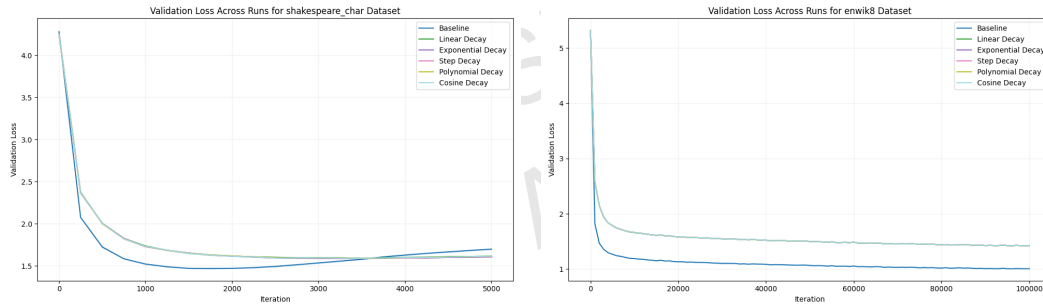
(a) Training loss for shakespeare_char

(b) Training loss for enwik8



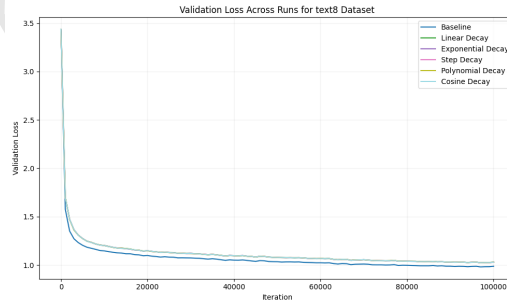
(c) Training loss for text8

Figure 1: Training loss over iterations for different datasets.



(a) Validation loss for shakespeare_char

(b) Validation loss for enwik8



(c) Validation loss for text8

Figure 2: Validation loss over iterations for different datasets.

7 CONCLUSIONS AND FUTURE WORK

In this paper, we investigated the implementation of layer-wise learning rates in transformer models to optimize training dynamics. By assigning unique learning rates to each layer, with deeper layers having lower learning rates, we aimed to enhance the training process. We modified the `configure_optimizers` function to support this approach and validated our method through extensive experiments on the `shakespeare_char`, `enwik8`, and `text8` datasets.

Our experimental results showed that the baseline model, which used a single learning rate for all layers, generally outperformed the models with layer-wise learning rates in terms of final training and validation losses. The training time and average inference tokens per second were comparable across different learning rate decay strategies. One possible reason for the higher losses in models with layer-wise learning rates is that the predefined learning rate decay strategies may not have been optimal for the specific datasets and model architecture used.

A limitation of our method is the use of predefined learning rate decay strategies, which may not be the best fit for all datasets and model architectures. Future work could explore adaptive learning rate decay strategies that dynamically adjust the learning rates based on the training dynamics. Additionally, further experiments with different model architectures and larger datasets could provide more insights into the effectiveness of layer-wise learning rates.

In future research, we plan to extend the application of layer-wise learning rates to other types of neural networks and investigate additional optimization techniques to enhance training efficiency and performance. We also aim to develop adaptive learning rate decay strategies that can dynamically adjust the learning rates based on the training dynamics, potentially leading to better performance and faster convergence.

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